

Team Number: 44

Project Title: Breast Cancer Stage Classification with miRNA

Report Date: 10/20/25

Part one: Progress status as a team

All team members have detailed tasks listed on Teams Planner: YES or NO; if NO, explain.

Yes

Please answer the following questions:

1. Have you met as a team this past week? If yes, give date/time, and the members attended the meeting. If no, explain.

No, we have not; however, we communicate almost daily through our team's Discord channel to align on project progress as well as updates.
2. Have you met with the sponsor as a team? If yes, give date/time, and the member attended the meeting. If no, explain.

No, we sent multiple emails in the past two weeks to get an update and feedback on our project update so we can progress to other tasks, but he hasn't been so responsive nor provided feedback to our updated tasks. Regardless, we took the initiative to make some progress with our project.

3. Describe verbally the tasks completed the past week and the challenges faced.

During Weeks 10 and 11, our team focused on finishing the stage-wise feature selection and analysis. We kept the setup consistent across all stages by always comparing Normal vs. Stage I, II, III, and IV with the same split, scaling, and parameters so results could be compared fairly. We re-ran all 3 feature selection approaches (traditional, deep learning, and clustering). For the traditional and deep learning methods, we generated Top-5, Top-10, and Top-15 miRNA lists for each stage. For clustering, we generated 2 clusters per stage using standardized miRNA expression data (features as samples) and then treated the smaller cluster as the stage-specific signal under the assumption that broadly expressed miRNAs form the large/background cluster and stage-enriched miRNAs separate into a minority cluster; if that cluster had fewer than 10 miRNAs, we treated it as noise and did not use it. Each team member worked on specific methods: Joshua ran LASSO, LSTM-Seq, and Gaussian Mixture; Anika handled SFS, FNN, and K-Means; Nicholas ran Chi-square, 1D-CNN, and Hierarchical; and Brehon handled Variance Threshold, DAE, and Spectral.

After the feature selection runs, Anika combined everyone's outputs into stage-wise consensus miRNA lists (traditional, deep learning, and clustering) and then used those selected features to train and test 4 classifiers (Logistic Regression, SVM, Random Forest, and KNN) for Normal vs. Stage I-IV. She recorded metrics such as accuracy, sensitivity, specificity, and AUC for each stage to summarize which stage-wise panels were most reliable and where performance was limited by data.

4. Describe the tasks to be completed the coming week.

Our team will finish preparing our project poster for the Senior Design Showcase, summarizing the overall workflow, main findings, and visual results from each phase of our analysis. Additionally, we will complete the Project Status Report slides for the second prototype demo, summarizing our project's goals, the progress made so far, and the key outcomes from each phase of the feature selection and classification process. In this demonstration, we will discuss how the combined feature-selection work supports the development of our breast cancer stage classification system and outline the next steps toward completion. In parallel, we will continue working on explainability by interpreting and validating the classification results now that model training is complete. We will also begin generating cleaned notebooks for each feature selection family (traditional, deep learning, and clustering) so that each stage-wise result can be reviewed independently and shared in a reproducible way.

Previous Report Summary. Date: 10/06/25

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
Joshua Adegboyo	2	0	3
Anika Baro Villa	5	0	3
Nicholas Campos	2	0	3
Brehon Parker	2	0	3

Current Report Summary. Date: 10/20/25

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
Joshua Adegboyo	3	0	2
Anika Baro Villa	5	0	3
Nicholas Campos	3	0	
Brehon Parker	3	0	2

Name of the Member: **Joshua Adegboyo**

Report Date: **10/20/25**

Task 8.1: Prepare LSTM Feature Outputs for Deep Learning Overlap Analysis (10/13/25)

This week, I contributed to the deep learning feature selection phase by finalizing my LSTM-Seq based miRNA feature extraction. My original LSTM model produced a list of top 10 significant miRNAs but based on group feedback and Dr. Ali’s updated direction to improve feature overlap across deep learning methods, I expanded my results by increasing k to 15 and 20.

These lists were shared with the team and used in the deep learning consensus overlap alongside features from FNN, CNN, and DAE. This step ensured my contribution was included in the final overlap comparison required before classification.

Task 8.2: Generate Final Deep Learning Feature Overlap List (10/14/25)

After preparing my LSTM feature lists, I verified overlap of my LSTM-selected miRNAs with other deep learning methods. The overlap list contained a small number of shared features, which confirmed what was discussed in the team meetings—that deep learning-based feature selection methods capture diverse signals from the data. My updated LSTM outputs (k=15, k=20) were submitted to support the final deep learning overlap file.

Task 9.1: Apply KNN Classification on Selected Feature Groups (10/18/25)

Following Anika’s instructions, I applied **K-Nearest Neighbors (KNN)** classification using the **70/30 train-test split** across all three required feature groups:

Feature Group	Input File Used
Traditional consensus features	consensus_traditional_atleast_2.csv
Deep learning consensus features	consensus_dl_atleast_2.csv
Clustering consensus features	consensus_clustering_atleast_2.csv

I also included my LSTM k15 and LSTM k20 feature sets for comparison. StandardScaler was applied before KNN training to normalize feature values. I used k = 5 neighbors, consistent with standard KNN baseline testing.

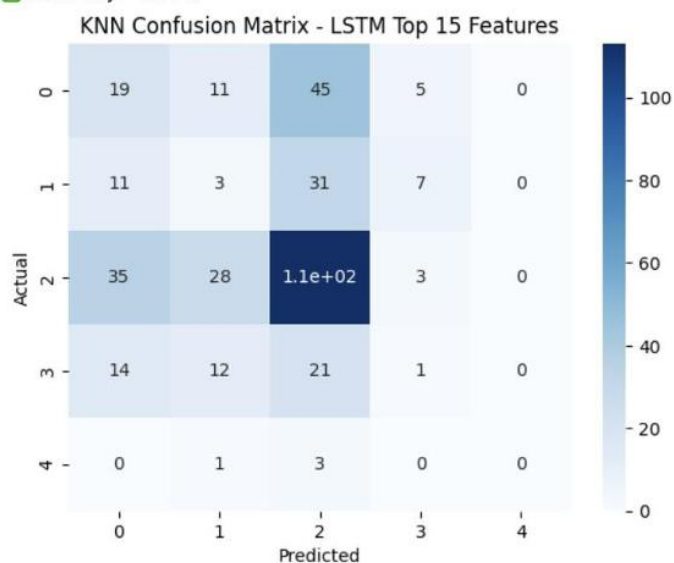
Results Summary (KNN Accuracy):

Feature Set	Accuracy (KNN)
LSTM k=15	~0.37–0.39
LSTM k=20	~0.39–0.40
Traditional Consensus	~0.36
Deep Learning Consensus	~0.35
Clustering Consensus	~0.38 (Best)

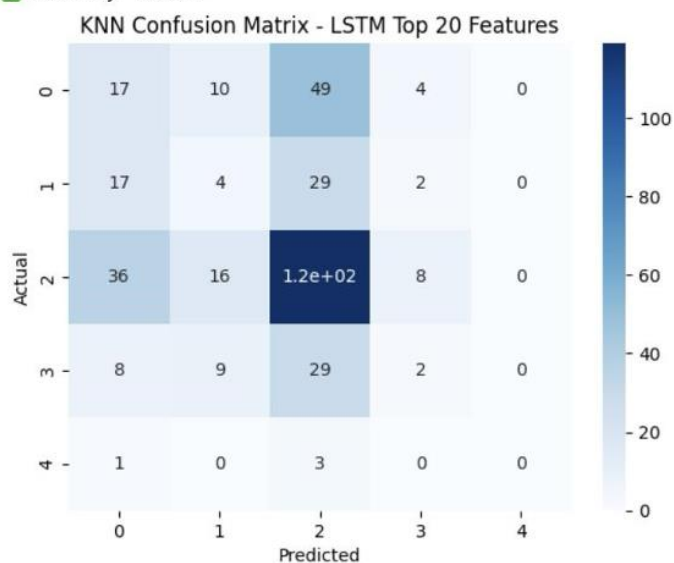
Confusion matrices were also generated for each trial to analyze misclassification patterns across breast cancer stages. As expected, Stage 4 showed the lowest classification performance due to class imbalance.

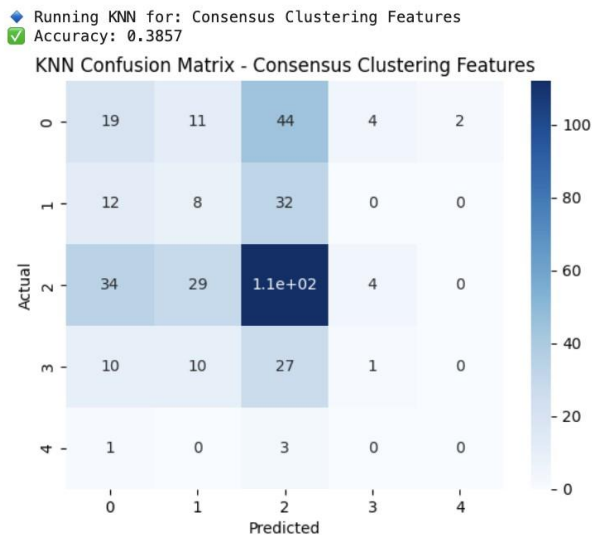
Features loaded:
LSTM k15: 15 features
LSTM k20: 20 features
Traditional: 14 features
Deep Learning: 3 features
Clustering: 116 features

Running KNN for: LSTM Top 15 Features
Accuracy: 0.3747



Running KNN for: LSTM Top 20 Features
Accuracy: 0.3912





Task 10.1: Apply Stage-Wise Traditional Feature Selection Using LASSO (10/25/25)

For this task, I implemented LASSO (Logistic Regression with L1 regularization) as my traditional feature-selection method. The procedure was performed stage by stage (Normal vs Stage I–IV) to identify the most influential miRNAs associated with each cancer stage. To maintain consistency with the deep-learning pipeline, each run followed by the same 70/30 train-test split, StandardScaler normalization, and class balancing approach. For each binary comparison, I trained multiple LASSO models across different C values and selected the best-performing one based on validation accuracy. Features were ranked according to the absolute coefficient magnitude of the best model. For every stage, I exported the Top 5, Top 10, and Top 15 miRNAs as separate CSV files alongside a full ranking file. These outputs will be used in the upcoming cross-method overlap analysis to compare traditional vs deep learning vs clustering results. Preliminary validation accuracies ranged between 0.61 and 0.67 across stages, confirming the method’s stability and reproducibility.

LASSO stage-wise summary:

	stage	val_accuracy	best_C	nonzero_coeffs	top3_preview
0	1	0.5682	0.20	130	hsa-mir-7111, hsa-mir-744, hsa-mir-598
1	2	0.5388	0.10	153	hsa-mir-1322, hsa-mir-1225, hsa-mir-6822
2	3	0.6016	0.05	3	hsa-mir-1273c, hsa-mir-7641-2, hsa-mir-6087
3	4	0.8810	2.00	69	hsa-mir-4773-2, hsa-mir-6867, hsa-mir-623

Task 10.2: Apply Stage-Wise Deep Learning Feature Selection Using LSTM (10/25/25)

As my deep-learning approach, I extended the stage-wise framework using a **Long Short-Term Memory (LSTM)** network. Each model compared Normal vs Stage I–IV samples separately, treating each miRNA as a sequential input feature. The LSTM architecture consisted of one LSTM layer (32 units), dense layers with ReLU activation, dropout for regularization, and an output sigmoid node for binary classification. Training used class-balanced weights and early stopping to prevent overfitting. After training, I calculated permutation importance scores for all miRNAs on the validation set to determine their impact on accuracy. For each stage, miRNAs were ranked by importance, and the Top 5, Top 10, and Top 15 miRNAs were exported as CSV files. Model validation accuracies ranged from

approximately 0.58 to 0.64 across stages, indicating meaningful pattern recognition despite limited sample sizes. These results confirm that the LSTM method can identify stage-specific non-linear relationships among miRNAs that traditional statistical methods may overlook.

```
Loaded: (1209, 1882), miRNAs=1881

=== Stage 1: 0 vs 1 ===
Val accuracy: 0.6136 | features considered: 300
Saved: stage 1 top5/10/15 + full_ranking

=== Stage 2: 0 vs 2 ===
Val accuracy: 0.6434 | features considered: 300
Saved: stage 2 top5/10/15 + full_ranking

=== Stage 3: 0 vs 3 ===
Val accuracy: 0.5312 | features considered: 300
Saved: stage 3 top5/10/15 + full_ranking

=== Stage 4: 0 vs 4 ===
Val accuracy: 0.8333 | features considered: 300
Saved: stage 4 top5/10/15 + full_ranking
```

Task 10.3: Apply Stage-Wise Clustering as Feature Selection Using K-Means (10/25/25)

The final phase of my feature selection work will apply K-Means clustering to detect groups of miRNAs with similar expression patterns for each stage comparison (Normal vs Stage I–IV). Each analysis will produce two clusters per stage, then we will determine the important cluster which may be determined either by literature support or by selecting the smaller cluster based on variance.

All preprocessing steps (StandardScaler normalization and data transposition) will be identical to the previous tasks to maintain comparability across methods. Outputs will include a pair of CSV files per stage (representing the two clusters) to support the upcoming cross-method overlap analysis scheduled for completion before poster submission on Nov 3.

Task 11.1: Generate Stage-Wise Consensus miRNA Lists (11/05/25)

For this task, I will integrate the outputs from my three feature-selection pipelines; LASSO (Traditional), LSTM (Deep Learning), and K-Means (Clustering) to construct unified, stage-specific consensus miRNA lists.

For each stage comparison (Normal vs Stage I–IV), I will collect the Top 5, Top 10, and Top 15 miRNA lists produced by all three methods. Using these sets, I will identify overlapping or consistently recurring miRNAs across methods. The degree of overlap will be used to determine which miRNAs show cross-method agreement, reflecting biological robustness and computational consistency. These files will represent the most reliable stage-specific biomarkers, serving as the foundation for the next classification phase.

Task 11.2: Train and Evaluate Stage-Wise Classifiers (11/09/25)

Using the consensus miRNA lists from Task 11.1, I will perform stage-wise classification to assess the discriminative capability of the selected biomarkers. Each classifier will focus on differentiating Normal vs Stage I–IV, maintaining the same 70/30 train-test split used in earlier phases.

I will train multiple machine-learning models; K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest (RF), and Logistic Regression (LR) to evaluate classification accuracy across methods. For each model and stage, I will compute key performance metrics, including accuracy, precision, recall, F1-score, and confusion

matrices. The results will be summarized in both tabular and graphical formats to compare classifier effectiveness per stage and determine which consensus-derived feature set yields the highest predictive performance.

Name of the Member: Anika Baro Villa

Report Date: 10/20/25

Task 8.1: Generate a list of overlapping miRNAs across different traditional feature selection methods (10/07/25)

For this task, I compared the sets of significant miRNAs identified by each traditional feature selection method we used (Chi2, LASSO, SFS, and VT) to find shared candidates and build a consensus list. The results showed that Chi2 and VT had the highest overlap (8 miRNAs), followed by Chi2 and LASSO (6), LASSO and VT (6), and LASSO and SFS (4), while no miRNAs were common across all 4 methods. I then computed a frequency-based consensus, finding that 14 miRNAs appeared in at least 2 methods, 5 in at least 3 methods, and none in all 4.

Task 8.2: Expanding FNN feature selection ($k = 15, 20$) to improve cross-method overlap (10/07/25)

I increased the number of selected miRNAs (k) in the Feed-Forward Neural Network (FNN) I implemented earlier. Previously, I used k values of 3, 5, 8, and 10, so this time I increased k to 15 and 20 to test whether the minimal overlap observed across our deep learning methods was due to overly restrictive cutoff criteria. After training the FNN with L1-regularized input weights and class-balanced loss, I ranked all 1,881 miRNAs by their average absolute weight importance and extracted the top- k subsets. I then evaluated these subsets using a logistic regression classifier with 3-fold cross-validation. I saved the ranked lists and corresponding performance metrics for downstream comparison with the 1D-CNN, AE, and LSTM-Seq results to assess consistency and potential improvements in cross-method overlap.

Task 8.3: Generate a list of overlapping miRNAs across different deep learning feature selection methods (10/10/25)

For this task, I generated a list of overlapping miRNAs across the deep learning methods (FNN, CNN, LSTM, and DAE) we used for feature selection. I performed the overlap analysis for both $k = 15$ and $k = 20$ to evaluate how the number of selected features affected consistency across methods. For $k = 15$, the overlap was minimal, with only 1 shared miRNA detected, while increasing k to 20 resulted in a slightly higher overlap between methods. No miRNAs were common to all 4 models in either case. I also computed frequency-based consensus lists, which showed 1 miRNA appearing in at least 2 methods for $k = 15$ and 3 for $k = 20$. I exported all results as CSV files for future cross-method and biological comparison.

Task 9.1: Apply Random Forest classification on miRNAs selected by feature selection approaches (10/15/25)

I applied Random Forest classification to the sets of miRNAs that were selected using our 3 main feature selection approaches: traditional methods, deep learning, and clustering. For each approach, I used the same preprocessing and evaluation pipeline to maintain consistency across all experiments. This pipeline included applying a $\log_{1p}$ transformation to the data and training a Random Forest classifier using cross-validation. I used accuracy as the main evaluation metric, as required by Dr. Ali Ibrahim, and recorded additional metrics such as macro-F1 and per-class recall to better understand how the model performed across all breast cancer stages. To ensure the results were reliable and not affected by data leakage, all transformations and model training were done within each cross-validation fold. After completing the classification for each of the 3 feature selection approaches, I summarized the mean and standard deviation of the cross-validated accuracy values to compare the model's performance when using miRNAs selected by different methods.

Task 9.2: Compare classification performance between traditional, DL, and clustering methods (10/16/25)

I compared the classification performance of the Random Forest models I created across the traditional, deep learning, and clustering-based feature selection approaches. The clustering-derived feature set contained 116 miRNAs and achieved the highest cross-validated accuracy of approximately 0.47 with the lowest variability across folds. The traditional feature set included 14 miRNAs and reached an accuracy of about 0.41, showing stable but

moderate performance. The deep learning consensus set had only 3 overlapping miRNAs and achieved a lower accuracy of around 0.25, which was expected given the small number of features and the difficulty of the five-class stage classification task. In all cases, accuracy was used as the primary metric for comparison, while macro-F1 and per-class recall were reviewed to understand class imbalance effects. All models showed low recall for stage 4 samples, indicating that additional strategies may be needed to improve performance on minority classes. Overall, this task helped me evaluate how well Random Forest performed across different sets of miRNAs and highlighted that the clustering-derived features provided the strongest predictive signal among the 3 approaches.

Task 10.1: Apply stage-wise traditional feature selection using SFS (10/22/25)

Our dataset contains miRNA expression profiles from 2 types of samples per comparison: non-cancer control tissue (Normal) and tumor tissue labeled by clinical stage (I-IV). So in this task, I treated each stage as its own problem (Normal vs. Stage I, Normal vs. Stage II, and so on) so that the results would be directly comparable across stages under one fixed setup and label scheme. I standardized the features and trained a logistic-regression model, then used Sequential Feature Selection, which builds the set one feature at a time (starting from none, testing each candidate, keeping the best, and repeating) while scoring with five-fold cross-validated ROC-AUC to keep the ranking robust and repeatable. For every stage, I generated 3 subsets (the top 5, 10, and 15 miRNAs) and saved them as CSV files for later comparison. Additionally, I printed accuracy and AUC as quick checks to document what the model saw, while keeping the actual selections strictly driven by the SFS procedure. The screenshots below show the SFS pipeline I ran per stage, and a sample output for Stage I, including the selected top 15 miRNAs and the train/test performance summary that I logged for traceability.

```
# For each k, run SFS (scaler inside CV via Pipeline)
for k in k_values:
    clf = Pipeline([
        ("scaler", StandardScaler()),
        ("lr", make_lr())
    ])
    sfs = SequentialFeatureSelector(
        estimator=clf,
        n_features_to_select=k,
        direction="forward",      # follow simple forward SFS
        scoring="roc_auc",
        cv=5,
        n_jobs=-1
    )
    sfs.fit(X_train, y_train)
```

```
Top 15 (SFS) miRNAs for Stage I:
1. hsa-mir-1284
2. hsa-mir-4310
3. hsa-mir-448
4. hsa-mir-4506
5. hsa-mir-4693
6. hsa-mir-4708
7. hsa-mir-4770
8. hsa-mir-4777
9. hsa-mir-548s
10. hsa-mir-615
11. hsa-mir-6750
12. hsa-mir-6768
13. hsa-mir-6835
14. hsa-mir-7111
15. hsa-mir-8059

Class distribution : Normal=265, Stage I=173
Split              : Train= 306, Test= 132
Accuracy           : Train=0.742 | Test=0.568
AUC                : Train=0.815 | Test=0.558 | Gap=0.257
Note               : metrics are sanity checks only; selection is SFS-driven
```

Task 10.2: Apply stage-wise deep learning feature selection using FNN (10/23/25)

Building directly on the SFS setup, I kept the same stage-wise framing (Normal vs. Stage I–IV) and fixed configuration so the deep-learning results are directly comparable. For each stage, I split the data with a smaller validation/test when the positive class is tiny, standardized features on the training set only, and computed balanced class weights. I then trained a small feed-forward neural network with 1 hidden layer (ReLU) plus dropout and L1+L2 regularization, optimized with Adam and early stopping to prevent overfitting. To stabilize the ranking, I repeated training across several fixed random seeds and, for each run, extracted per-miRNA importance from the first dense layer by summing the absolute input-to-hidden weights; I averaged these importances across seeds and sorted the features. For every stage, I saved the full ranking to CSV and exported 3 subsets (Top-5, Top-10, Top-15) for downstream comparison; accuracy and AUC were printed as sanity checks only, since selection is weight-based. The screenshots below show the FNN architecture I used and a sample output for Stage I, including the per-stage data split, sanity-check metrics, and the Top-5 miRNAs produced.

```
# FNN with regularization to prevent overfitting
def build_model(input_dim: int) -> Sequential:
    model = Sequential([
        Input(shape=(input_dim,)),
        Dense(16, activation='relu', kernel_regularizer=l1_l2(l1=0.001, l2=0.001)),
        Dropout(0.5),
        Dense(1, activation='sigmoid')
    ])
    model.compile(
        optimizer=tf.keras.optimizers.Adam(learning_rate=0.0001),
        loss='binary_crossentropy',
        metrics=['accuracy']
    )
    return model
```

```
=====
Stage I
=====
Class distribution : Normal=265, Stage I=173
Split              : Train= 306, Test= 132
Accuracy           : Train=0.912 | Test=0.629
AUC                : Train=0.952 | Test=0.550 | Gap=0.403
These metrics are sanity checks only; ranking is weight-based

FNN-identified miRNA signatures

Top 5 miRNAs for Stage I:
1. hsa-mir-5703
2. hsa-mir-4652
3. hsa-mir-4773-1
4. hsa-mir-6808
5. hsa-mir-3199-1
```

Task 10.3: Apply stage-wise clustering as feature selection using K-Means (10/24/25)

I kept the same stage-wise framing used earlier, treating each comparison separately as Normal versus Stage I, II, III, and IV so results are directly comparable. For each pairing, I filtered the data to those 2 labels, separated expression from labels, transposed into miRNA by samples, and standardized before clustering. I then ran K-Means with 2 clusters and recorded the silhouette score and cluster sizes to document separation quality. After fitting, I saved the miRNA membership for each cluster to CSV files and produced a simple PCA plot to visualize the 2 clusters in the first 2 principal components. Finally, I summarized each stage's silhouette and cluster counts in a small table to support downstream selection and overlap work. The screenshots below show the K-Means clustering code I used to generate

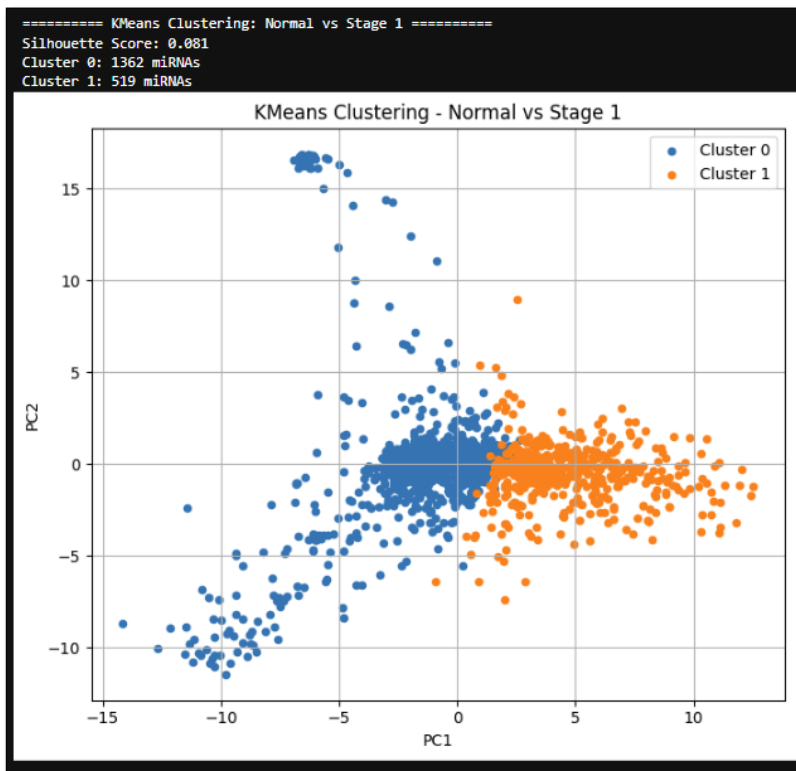
per-stage miRNA clusters and an example Stage I output, including the silhouette score, the sizes of the 2 clusters, and the PCA visualization of those clusters.

```
# Apply K-Means clustering
cluster_model = KMeans(n_clusters=2, random_state=42, n_init="auto")
cluster_labels = cluster_model.fit_predict(data_T_scaled)

# Silhouette score
sil = silhouette_score(data_T_scaled, cluster_labels)
print(f"Silhouette Score: {sil:.3f}")

# Cluster sizes
unique, counts = np.unique(cluster_labels, return_counts=True)
cluster_sizes = dict(zip(unique, counts))
for c, size in cluster_sizes.items():
    print(f"Cluster {c}: {size} miRNAs")

# Save miRNAs by cluster
clusters = pd.DataFrame({"miRNA": data_T.index, "cluster": cluster_labels}).set_index("miRNA")
for c in [0, 1]:
    cluster_mirnas = clusters[clusters["cluster"] == c].index
    cluster_mirnas.to_series().to_csv(
        f"stage({stage_code})_cluster({c})_miRNAs.csv", index=False, header=["miRNA"]
    )
```



Task 11.1: Generate stage-wise consensus miRNA lists per feature-selection approach (10/30/25)

For this task, I combined the outputs from all individual feature selection methods to generate unified, stage-specific consensus miRNA lists for each family (traditional, DL, and clustering). Before running the consensus script, I validated that all expected input files were present, correctly formatted, and contained no missing or duplicate entries. Using these verified files, I built frequency tables showing how often each miRNA was selected across the 4 methods in its family and then computed 3 consensus sets per stage and k: intersection (miRNAs selected by all 4 methods), majority (selected by ≥ 2 methods), and union (selected by any method). I saved these consensus lists for each stage and approach, along with summary tables detailing the number of methods contributing to each set. The screenshot below shows the summary output displaying the number of intersecting, majority, and union miRNAs for each stage and approach.

Approach	Stage	k	n_methods_found	n_intersection	n_majority	n_union
Traditional	Stage I	5	4	0	4	16
Traditional	Stage I	10	4	0	6	34
Traditional	Stage I	15	4	0	8	51
Traditional	Stage II	5	4	0	3	16
Traditional	Stage II	10	4	0	6	33
Traditional	Stage II	15	4	0	9	49
Traditional	Stage III	5	4	0	2	18
Traditional	Stage III	10	4	0	7	33
Traditional	Stage III	15	4	0	9	50
Traditional	Stage IV	5	4	0	3	17
Traditional	Stage IV	10	4	0	6	34
Traditional	Stage IV	15	4	0	6	53
DeepLearning	Stage I	5	4	0	0	20
DeepLearning	Stage I	10	4	0	1	39
DeepLearning	Stage I	15	4	0	2	58
DeepLearning	Stage II	5	4	0	0	20
DeepLearning	Stage II	10	4	0	0	40
DeepLearning	Stage II	15	4	0	1	59
DeepLearning	Stage III	5	4	0	0	20
DeepLearning	Stage III	10	4	0	0	40
DeepLearning	Stage III	15	4	0	0	60
DeepLearning	Stage IV	5	4	0	1	19
DeepLearning	Stage IV	10	4	0	1	39
DeepLearning	Stage IV	15	4	0	1	59
Clustering	Stage I		4	0	594	646
Clustering	Stage II		4	0	98	664
Clustering	Stage III		4	0	600	691
Clustering	Stage IV		4	0	89	800

Task 11.2: Train and evaluate stage-wise classifiers (10/30/25)

For each stage (Normal vs. Stage I-IV), I used the consensus miRNA lists generated and used them as input features to train 4 classifiers: Logistic Regression, linear SVM, Random Forest, and KNN. I run all models on a consistent train/test split per stage, with scaling handled inside the pipeline and class weights turned on for imbalance. For every (stage, feature set, classifier) combination I recorded accuracy, sensitivity, specificity, and AUC, and saved all results to a csv file. As expected, Stages I-III produced meaningful signal, while Stage IV remained unstable because it only has 15 breast cancer samples total. The screenshot below shows a sample output for Stage I, listing each classifier, feature-selection method, and its corresponding performance metrics.

Stage I						
Classifier	Method	Accuracy	Sensitivity	Specificity	AUC	
KNN	Clustering-majority	0.470	0.615	0.375	0.495	
KNN	DeepLearning-top10-majority	0.583	0.038	0.938	0.475	
KNN	DeepLearning-top15-majority	0.523	0.154	0.762	0.449	
KNN	Traditional-top10-majority	0.530	0.327	0.662	0.465	
KNN	Traditional-top15-majority	0.523	0.288	0.675	0.481	
KNN	Traditional-top5-majority	0.523	0.346	0.638	0.518	
LR	Clustering-majority	0.598	0.423	0.712	0.569	
LR	DeepLearning-top10-majority	0.424	0.808	0.175	0.490	
LR	DeepLearning-top15-majority	0.492	0.615	0.412	0.472	
LR	Traditional-top10-majority	0.538	0.538	0.538	0.539	
LR	Traditional-top15-majority	0.530	0.481	0.562	0.526	
LR	Traditional-top5-majority	0.568	0.462	0.638	0.541	
RF	Clustering-majority	0.598	0.096	0.925	0.493	
RF	DeepLearning-top10-majority	0.394	0.885	0.075	0.473	
RF	DeepLearning-top15-majority	0.523	0.250	0.700	0.440	
RF	Traditional-top10-majority	0.545	0.365	0.662	0.524	
RF	Traditional-top15-majority	0.530	0.385	0.625	0.524	
RF	Traditional-top5-majority	0.538	0.385	0.638	0.536	
SVM-linear	Clustering-majority	0.515	0.346	0.625	0.453	
SVM-linear	DeepLearning-top10-majority	0.402	0.923	0.062	0.510	
SVM-linear	DeepLearning-top15-majority	0.386	0.865	0.075	0.525	
SVM-linear	Traditional-top10-majority	0.553	0.500	0.588	0.545	
SVM-linear	Traditional-top15-majority	0.523	0.481	0.550	0.535	
SVM-linear	Traditional-top5-majority	0.614	0.404	0.750	0.538	

Task 12.1: Prepare project poster for senior design showcase (11/03/25)

I will finish working on our project poster. I am responsible for creating the header, problem statement, and workflow diagram, making sure the layout clearly presents our stage-wise approach and main results. I will design the workflow diagram to show the full pipeline from data preprocessing through feature selection, consensus, classification, and explainability/literature validation. I will also review the remaining sections to align terminology and formatting so our poster presents our stage-wise feature selection research and key results coherently.

Task 12.2: Package stage-wise notebooks for traditional, deep Learning, and clustering feature selection (11/05/25)

For this task, I will clean and finalize the stage-wise notebooks for each feature selection approach we use in our breast cancer classification system (traditional, deep learning, and clustering). Each notebook will show the Normal vs. Stage I-IV setup, preprocessing steps, and selected miRNAs (Top-5/10/15 or cluster output) in a clear, shareable format. I will make sure the process is fully reproducible by keeping the code, parameters, and outputs consistent across all notebooks.

Task 12.3: Prepare second project status report slides (11/09/25)

I will prepare the demo slide section summarizing the current status of our project and the work we have completed so far. The slides for this section will highlight key milestones such as downloading and cleaning the dataset, linking clinical stage labels, and completing stage-wise feature selection across 3 feature selection approaches (traditional, deep learning, and clustering). I will also include the generation of stage-specific consensus miRNA panels, the completion of stage-wise classification using these panels, and the initial interpretation of results across methods and stages.

Name of the Member: **Nicholas Campos**

Report Date: **10/20/25**

Task 8.1: SVM Classification on Selected miRNA Features (10/9/25)

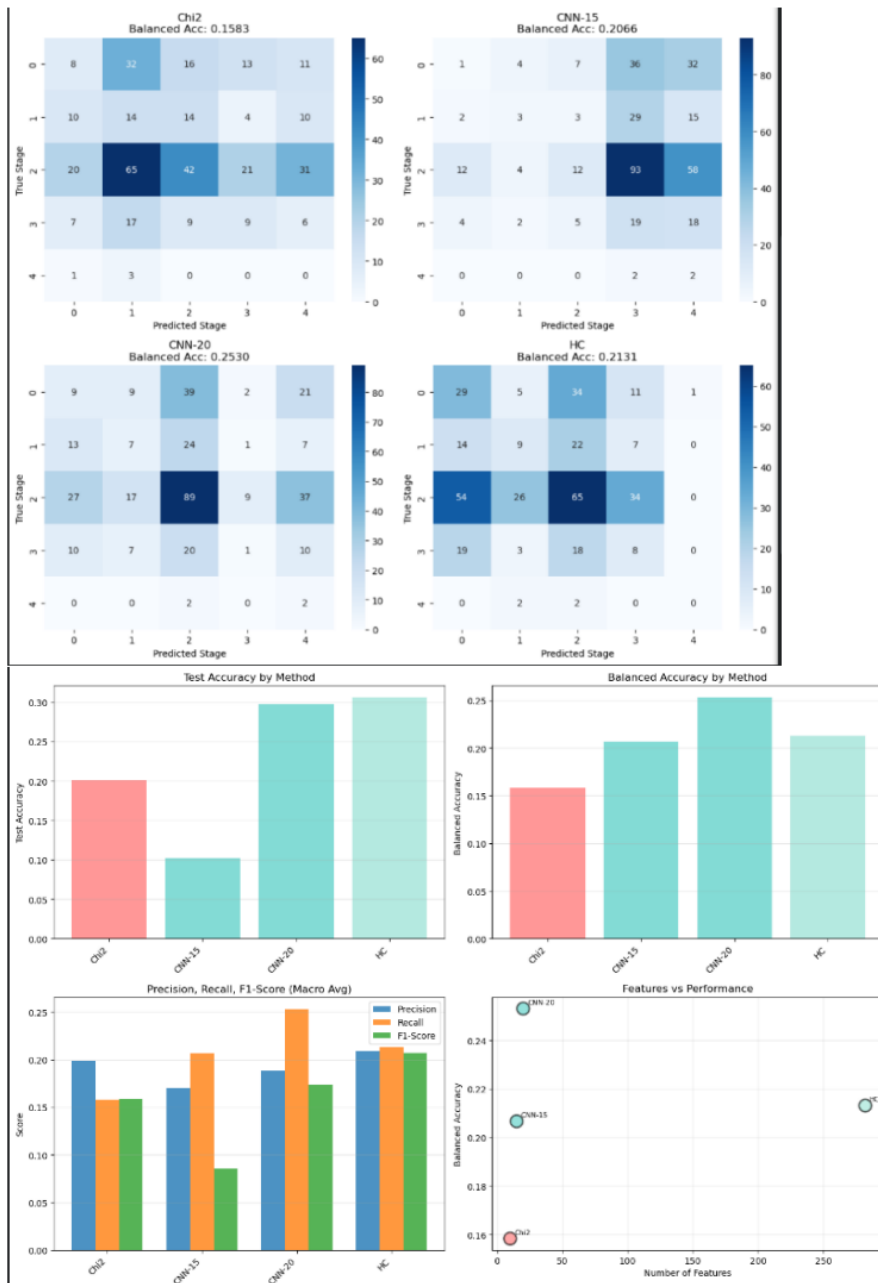
Following Dr. Ali's instructions, I implemented Support Vector Machine (SVM) classification to evaluate the miRNA features selected from three distinct feature selection approaches: traditional methods (Chi2), deep learning (1DCNN), and clustering (Hierarchical Clustering). For each approach, I loaded the corresponding important miRNA CSV files, extracted those specific features from the dataset, and trained separate SVM classifiers maintaining the same 70/30 data split protocol for consistency. The SVM models were trained with optimized hyperparameters to predict breast cancer stage classification (Stage I-IV). The classification results provided comprehensive performance metrics including accuracy, precision, recall, and F1-scores for each feature selection method. The deliverable includes documented accuracy metrics for all three approaches, confusion matrices showing classification performance across cancer stages, and trained SVM models ready for comparative analysis with other team members' classifiers.

```
=====
COMPREHENSIVE PERFORMANCE SUMMARY - ALL METHODS
=====
```

Method	Category	Features	Test Acc	Balanced Acc	Precision	Recall	F1-Score
Chi2	Traditional	10	0.2011	0.1583	0.1982	0.1583	0.1585
CNN-15	Deep Learning	15	0.1019	0.2066	0.1700	0.2066	0.0855
CNN-20	Deep Learning	20	0.2975	0.2530	0.1884	0.2530	0.1736
HC	Clustering	282	0.3058	0.2131	0.2089	0.2131	0.2072

```
🏆 BEST METHOD: CNN-20
  Balanced Accuracy: 0.2530
  Category: Deep Learning

=====
SAVING RESULTS TO FILES
=====
```



Task 8.2: Cross-Classifer Performance Comparison (10/9/25)

Building on my SVM classification results, I will collaborate with my team members to conduct a comprehensive comparison of classifier performance across different algorithms as requested by Dr. Ali. Each team member will apply a different classifier (KNN, Random Forest, Logistic Regression, and my SVM) to the same three sets of selected miRNA features from traditional, deep learning, and clustering methods. This systematic comparison will reveal which combination of feature selection method and classifier produces the best results for breast cancer stage classification. I will compile accuracy metrics from all team members' classifiers and create a comparative analysis matrix showing performance differences across methods. The analysis will identify the optimal feature selection approach by examining which method consistently produces high accuracy across multiple classifiers, and determine whether traditional statistical methods, deep learning-based selection, or unsupervised clustering yields superior features for cancer stage prediction. This comparison will directly address Dr. Ali's question about which

method gives better results and provides evidence-based recommendations for the final diagnostic system implementation.



10.1: Apply stage-wise traditional feature selection using Chi-Square (10/25/25)

In this task, I'll perform Chi-Square statistical analysis individually for each cancer stage comparison (Normal vs. Stage I, Normal vs. Stage II, etc.) to identify stage-specific important miRNA features. My approach maintains consistency by using StandardScaler for normalization and SVM with balanced class weighting as the classifier. For every stage comparison, I'll identify the Top-15 and Top-20 features with the highest chi-square statistical scores and export these to CSV format. Performance metrics will serve primarily to verify the feature selection process is working correctly.

```
=====
Stage I
=====
Class distribution : Normal=265, Stage I=173
Chi2 scores computed for all 1881 miRNAs
Top feature: hsa-mir-148a (score=114907.4216)

Chi2-identified miRNA signatures

Top 5 miRNAs for Stage I:
1. hsa-mir-148a
2. hsa-mir-21
3. hsa-mir-10a
4. hsa-mir-30a
5. hsa-mir-9-1

Top 10 miRNAs for Stage I:
1. hsa-mir-148a
2. hsa-mir-21
3. hsa-mir-10a
4. hsa-mir-30a
5. hsa-mir-9-1
6. hsa-mir-9-3
7. hsa-mir-9-2
8. hsa-mir-206
9. hsa-mir-142
10. hsa-mir-1-2

Top 15 miRNAs for Stage I:
1. hsa-mir-148a
2. hsa-mir-21
3. hsa-mir-10a
4. hsa-mir-30a
5. hsa-mir-9-1
6. hsa-mir-9-3
7. hsa-mir-9-2
8. hsa-mir-206
9. hsa-mir-142
10. hsa-mir-1-2
11. hsa-mir-1-1
12. hsa-mir-375
13. hsa-mir-143
14. hsa-mir-133a-1
15. hsa-mir-99b
```

10.2: Apply stage-wise deep learning feature selection using 1D-CNN (10/25/25)

I'll implement a 1D Convolutional Neural Network with an attention layer to perform stage-by-stage feature selection. The network architecture incorporates dropout layers and early stopping mechanisms to prevent overfitting during training. My workflow involves running separate models for each stage comparison (Normal vs. Stage I-IV), then extracting the Top-15 and Top-20 miRNAs by analyzing the learned convolutional weights and computing feature importance rankings. Accuracy and balanced accuracy metrics will be calculated to ensure the model training was successful.

```

=====
Stage I
=====
Class distribution : Normal=265, Stage I=173
Split              : Train= 306, Test= 132
Accuracy           : Train=0.598 | Test=0.606
AUC                : Train=0.499 | Test=0.466 | Gap=0.034
These metrics are sanity checks only; ranking is weight-based

1D-CNN-identified miRNA signatures

Top 5 miRNAs for Stage I:
 1. hsa-mir-124-1
 2. hsa-mir-506
 3. hsa-mir-5692a-2
 4. hsa-mir-548aq
 5. hsa-mir-645

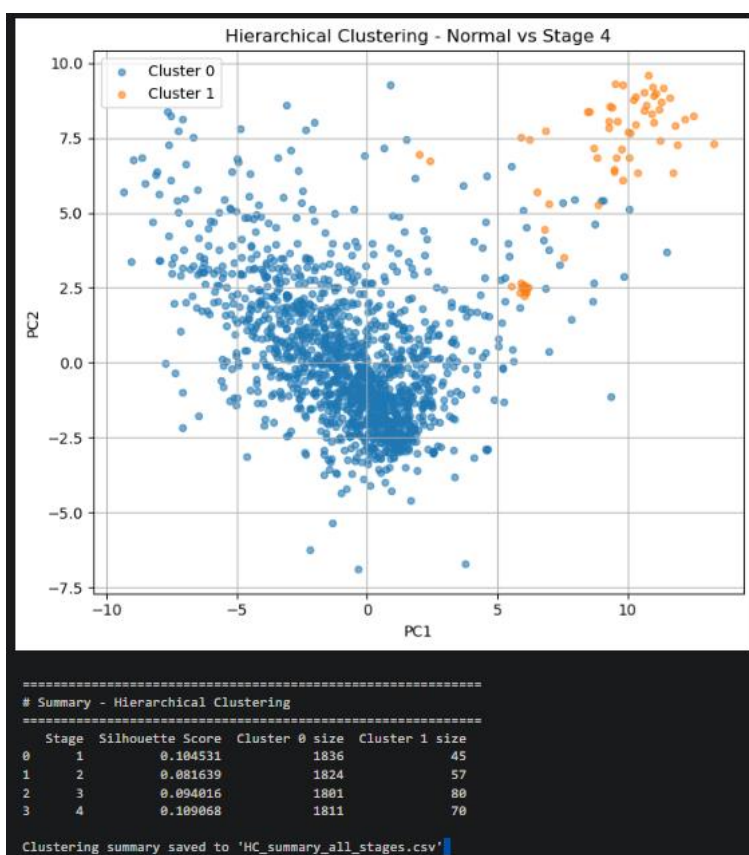
Top 10 miRNAs for Stage I:
 1. hsa-mir-124-1
 2. hsa-mir-506
 3. hsa-mir-5692a-2
 4. hsa-mir-548aq
 5. hsa-mir-645
 6. hsa-mir-499b
 7. hsa-mir-6760
 8. hsa-mir-509-2
 9. hsa-mir-512-1
10. hsa-mir-6861

Top 15 miRNAs for Stage I:
 1. hsa-mir-124-1
 2. hsa-mir-506
 3. hsa-mir-5692a-2
 4. hsa-mir-548aq
 5. hsa-mir-645
 6. hsa-mir-499b
 7. hsa-mir-6760
 8. hsa-mir-509-2
 9. hsa-mir-512-1
10. hsa-mir-6861
11. hsa-mir-514b
12. hsa-mir-1197
13. hsa-mir-640
14. hsa-mir-6816
15. hsa-mir-1203

```

10.3: Apply stage-wise clustering as feature selection using Hierarchical Clustering (10/25/25)

This task focuses on applying Hierarchical Clustering with Ward linkage to discover important features for each stage independently. I'll run separate clustering analyses comparing Normal samples against each cancer stage (I through IV), following consistent data preprocessing and transposition steps. Rather than using cluster membership alone, I'll apply a variance-based approach to select the most informative miRNAs—specifically, choosing the top 15% of features ranked by their variance across patient samples. Each stage's important features will be documented in individual CSV files.



Name of the Member: Brehon Parker

Report Date: 10/20/25

Task 7.1: Traditional (VT) and Deep Learning (DAE) Feature Outputs for Overlap Analysis (10/04/25)

Using both my Variance Threshold (traditional) and Denoising Autoencoder (deep learning) feature selection approaches, I prepared and saved the lists of significant miRNAs identified by each method. For Variance Threshold, I exported the selected features into standardized CSV outputs, formatted for consistency with the team. For DAE, I generated top-10 feature lists, ensuring they align with the simplified naming structure for later overlap comparison. These deliverables ensure that my results can be directly integrated with other team outputs (Chi2, CNN, clustering methods, etc.) to create overlap lists across traditional, deep learning, and clustering pipelines.

Task 7.2: Prepare project first demo slides (10/04/25)

For the first demo presentation, I worked on the dataset overview slide. I introduced the scope of our data — 1,209 patient samples and 1,881 miRNAs — to give context before diving into feature selection and classification. At first, I added bar and scatter plots, but after reviewing them with the team, I realized they didn't really add much value, so I removed them. Instead, I focused on keeping the slide clean, consistent with the rest of the deck, and easy to follow. My goal was to make sure the dataset was presented clearly without unnecessary clutter, so the audience can quickly understand the foundation of our work.

What has been done?



Data Preparation

- Data cleaned, standardized, and labeled with breast cancer stage information (I–IV)



Feature Selection Run

- Traditional: SFS, VT, Chi², LASSO
- Deep Learning: FNN, DAE, 1D-CNN, LSTM-Seq
- Clustering: KM, SC, GMM, HC



Overlap Lists

- Generated overlapping miRNA feature lists from traditional, deep learning, and clustering methods as input for classification



Literature Review

- Reviewed published studies and curated databases for miRNAs selected by our methods



Early Insights

- Overlaps across methods are limited, yet they highlight candidate features that may be informative for breast cancer stage classification



1,209 Samples
(Patients)



1,881 miRNAs
(Features)

Task 8.2: Apply Logistic Regression Classification on miRNAs Selected by Feature-Selection Approaches (10/17/25)

I re-ran my **Denoising Autoencoder (DAE)** model and expanded the number of selected features from $k = 10$ to $k = 15$ and $k = 20$ to increase overlap potential with other deep-learning methods. The training was performed on the breast_cancer(in).csv dataset, using Gaussian noise regularization and early stopping to prevent overfitting. After training, I ranked all 1,881 miRNAs using permutation-based importance and exported the top 15 and top 20 as separate files (dae_k15_features.csv, dae_k20_features.csv). These updated feature lists were shared with the team for integration into Anika's deep-learning overlap analysis.

k=15

```
miRNA
hsa-mir-323a
hsa-mir-323b
hsa-mir-431
hsa-mir-370
hsa-mir-485
hsa-mir-539
hsa-mir-329-1
hsa-mir-433
hsa-mir-432
hsa-mir-127
hsa-mir-382
hsa-mir-134
hsa-mir-136
hsa-mir-495
hsa-mir-493
```

k=20

```
miRNA
hsa-mir-323a
hsa-mir-323b
hsa-mir-431
hsa-mir-370
hsa-mir-485
hsa-mir-539
hsa-mir-329-1
hsa-mir-433
hsa-mir-432
hsa-mir-127
hsa-mir-382
hsa-mir-134
hsa-mir-136
hsa-mir-495
hsa-mir-493
hsa-mir-487b
hsa-mir-770
hsa-mir-668
hsa-mir-154
hsa-mir-1197
```

Task 9.1: Apply Logistic Regression Classification on miRNAs Selected by Feature Selection Approaches

(10/17/25)

I applied **Logistic Regression (LR)** classification using miRNAs selected by both my **Variance Threshold (VT)** and **DAE (k = 15, 20)** feature-selection methods. The pipeline included data standardization (StandardScaler), a 70/30 train-test split, and the LR model configured with balanced class weights and multinomial loss. Evaluation metrics included accuracy and macro-F1 score. The classifier achieved accuracies ranging from **0.33 to 0.37** and macro-F1 scores between **0.14 and 0.20** across the feature sets, with the DAE-derived features performing more consistently.

Used 10 features (ignored 0 missing).				
	precision	recall	f1-score	support
0	0.21	0.11	0.15	80
1	0.00	0.00	0.00	52
2	0.48	0.63	0.55	179
3	0.00	0.00	0.00	48
4	0.02	0.25	0.03	4
accuracy			0.34	363
macro avg	0.14	0.20	0.14	363
weighted avg	0.28	0.34	0.30	363
Accuracy: 0.339 Macro-F1: 0.144				
=== dae_k15_features.csv ===				
Used 15 features (ignored 0 missing).				
	precision	recall	f1-score	support
0	0.33	0.07	0.12	80
1	0.17	0.12	0.14	52
2	0.50	0.64	0.57	179
3	0.19	0.10	0.13	48
4	0.02	0.25	0.03	4
accuracy			0.37	363
macro avg	0.24	0.24	0.20	363
weighted avg	0.37	0.37	0.34	363
Accuracy: 0.366 Macro-F1: 0.199				
=== dae_k20_features.csv ===				
Used 20 features (ignored 0 missing).				
	precision	recall	f1-score	support
0	0.30	0.10	0.15	80
1	0.20	0.13	0.16	52
2	0.48	0.60	0.53	179
3	0.09	0.06	0.07	48
4	0.02	0.25	0.04	4
accuracy			0.35	363
macro avg	0.22	0.23	0.19	363
weighted avg	0.34	0.35	0.33	363
Accuracy: 0.347 Macro-F1: 0.192				
Feature file #Used #Missing Accuracy Macro-F1				
0	vt_significant.csv	10	0	0.339 0.144
1	dae_k15_features.csv	15	0	0.366 0.199
2	dae_k20_features.csv	20	0	0.347 0.192

Task 10.1: Apply Stage-Wise Traditional Feature Selection Using Variance Threshold (10/25/25)

I applied the **Variance Threshold (VT)** method individually for each cancer-stage comparison (Normal vs Stage I–IV) to identify the most informative miRNAs per stage. Using the same preprocessing pipeline as previous tasks (StandardScaler and train_test_split with a 70/30 split), I calculated feature variances and exported the Top-5, Top-10, and Top-15 features for each stage. The resulting CSV files (e.g., vt_stage_I_k5.csv, vt_stage_I_k10.csv, vt_stage_I_k15.csv) will be used alongside deep-learning and clustering methods during the stage-wise comparison and consensus generation phases.

Task 10.2 : Apply Stage-Wise Deep Learning Feature Selection Using DAE (10/25/25)

I extended my previous Denoising Autoencoder (DAE) workflow to a stage-wise setup, training independent autoencoders for each comparison (Normal vs Stage I–IV). Each DAE used Gaussian-noise regularization, dropout layers, and early stopping to prevent overfitting. Feature importance scores were derived via permutation error, and the Top-5, Top-10, and Top-15 miRNAs per stage were exported as individual CSVs (e.g., dae_stage_III_k10.csv). These files complete my deep-learning deliverables and will feed into Anika’s consensus and classification pipelines.

Task 10.3: Apply Stage-Wise Clustering as Feature Selection Using Spectral Clustering (10/25/25)

I implemented **Spectral Clustering** for each stage comparison (Normal vs Stage I–IV) using a two-cluster configuration (`n_clusters = 2`, `affinity = "rbf"`). Each subset was standardized with `StandardScaler`, and two clusters were generated per stage. For each stage, I exported two CSV files (`cluster0`, `cluster1`) and a `*_sizes.csv` summary showing cluster membership counts. Anika will select the “important” cluster for each stage during the consensus

Task 11.1: Generate Stage-Wise Consensus miRNA List

I will integrate outputs from the three feature-selection methods—Variance Threshold (VT), Denoising Autoencoder (DAE), and Spectral Clustering—to create unified consensus miRNA panels for each stage (I–IV). For each stage, I will compare the Top-5, Top-10, and Top-15 lists from all methods to identify overlapping or frequently recurring miRNAs. These will be compiled into consolidated consensus CSV files (e.g., `consensus_stage_I.csv` through `consensus_stage_IV.csv`), representing the most consistently selected miRNAs across multiple approaches. These consensus lists will serve as the foundation for classifier training and evaluation in the next phase.

Task 11.2: Train and Evaluate Stage-Wise Classifiers

Using the consensus miRNA sets, I will conduct stage-wise classification experiments to evaluate the discriminative power of the selected features. Each model will be trained on a 70/30 train-test split using Support Vector Machine (SVM) and Random Forest classifiers. Performance metrics—including accuracy, precision, recall, and F1-score—will be recorded for each stage to assess classification effectiveness. The resulting summary tables and confusion matrices will guide model refinement and support future explainability and literature validation efforts.